

Analyzing Toronto Apartment Rental Prices

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Project Task Assignments:

- Austin did...
- Olivia did...
- Raina did...
- Woobin did...

Introduction

This section should describe the background and objectives of the study.

Data Description

We obtained the data from [this](#) Kaggle dataset. The amount of rows in the original dataset before any data cleaning is 1,124. The data contains information on apartment rental units in Toronto collected in 2018, including the apartment's geographic information (`address`, `latitude`, and `longitude`), information on its rooms (the number of `bedrooms`, `bathrooms`, and `dens`), and the unit's monthly rental `price`.

The total number of rows in the original dataset is 1,124. There are no missing values in any columns.

During data cleaning, we removed 308 duplicates which likely occurred because the data came from multiple sources. We also removed 5 rows with unusual values for `price` (\$65/month, \$99/month, \$99/month, \$36,900/month, \$535,000/month). These outliers may skew our results so we decided to remove them. The sample size is now `n = 811`.

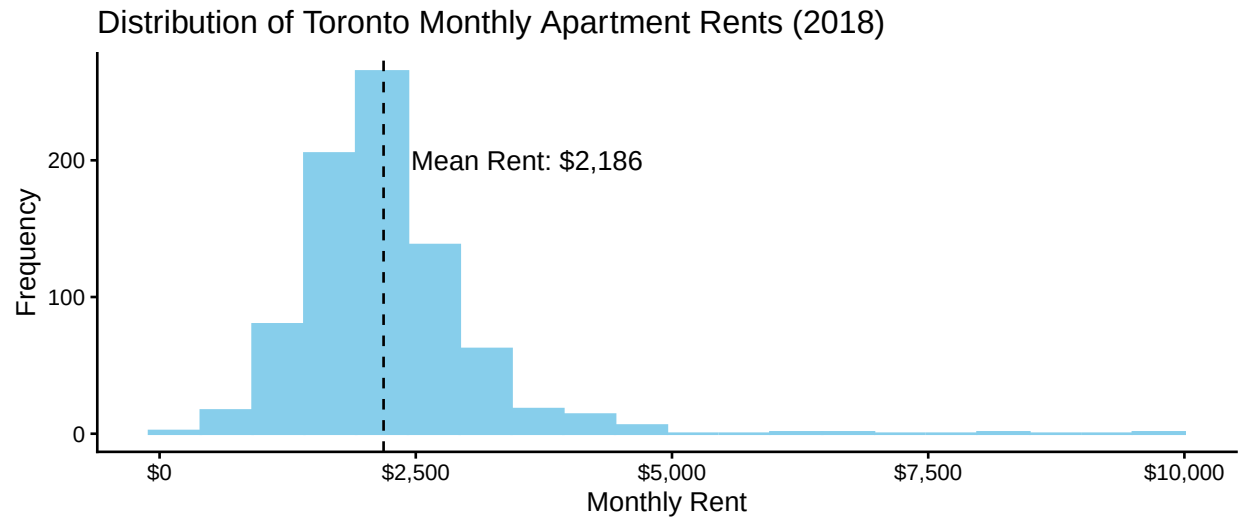
Exploratory Data Analysis

Before modeling, we want to find out more about the data. The variables we're most interested in are `price`, `bedroom`, `bathroom`, and `den`; however, we first view the sample's summary statistics of each column.

```
## # A tibble: 14 x 13
##   column      n    mean    sd  median trimmed  mad     min     max range
##   <chr>    <dbl>  <dbl> <dbl>  <dbl>   <dbl> <dbl>   <dbl>  <dbl> <dbl>
## 1 bedroom~    1    1.37    NA    1.37    1.37    0    1.37    1.37    0
## 2 bedroom~    1    0.567    NA    0.567    0.567    0    0.567    0.567    0
## 3 bathroo~    1    1.24    NA    1.24    1.24    0    1.24    1.24    0
## 4 bathroo~    1    0.438    NA    0.438    0.438    0    0.438    0.438    0
## 5 den_mean    1    0.154    NA    0.154    0.154    0    0.154    0.154    0
## 6 den_sd      1    0.361    NA    0.361    0.361    0    0.361    0.361    0
## 7 lat_mean    1   43.7    NA   43.7    43.7    0   43.7    43.7    0
## 8 lat_sd      1    0.685    NA    0.685    0.685    0    0.685    0.685    0
## 9 long_me~    1  -79.5    NA  -79.5  -79.5    0  -79.5   -79.5    0
##10 long_sd     1    1.84    NA    1.84    1.84    0    1.84    1.84    0
##11 price_m~    1 2186.    NA 2186.  2186.    0 2186.  2186.    0
##12 price_sd    1   787.    NA   787.   787.    0   787.   787.    0
##13 high_pr~    1    0.470    NA    0.470    0.470    0    0.470    0.470    0
##14 high_pr~    1    0.499    NA    0.499    0.499    0    0.499    0.499    0
## # i 3 more variables: skew <dbl>, kurtosis <dbl>, se <dbl>
```

TODO: a table of summary stats of everything except address

To visualize the distribution of `price`, we plot a histogram.

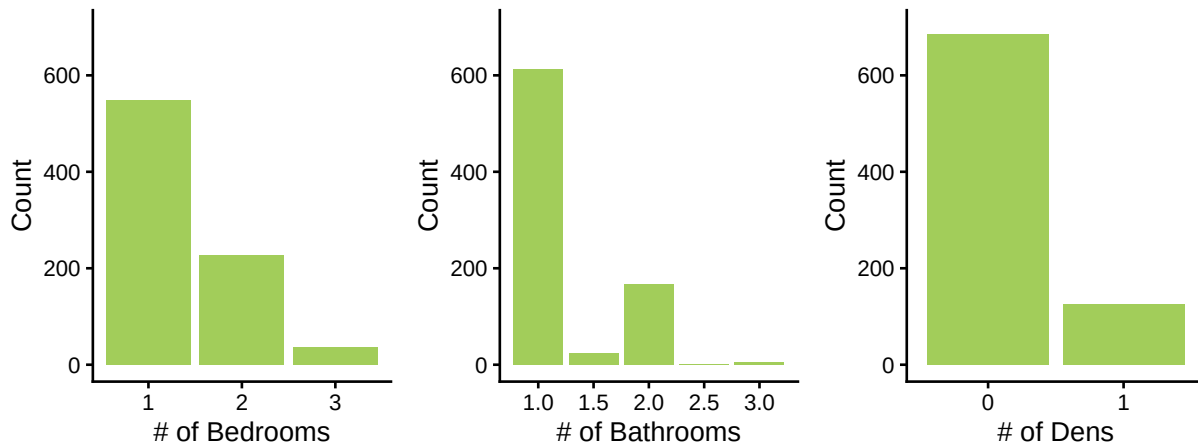


The distribution is slightly right-skewed and unimodal around the mean of `mean`. As well, its standard deviation is \$1,021/month, the minimum is \$150/month, and the maximum is \$9,750/month.

TODO: more commentary

Next, we view the frequencies of each discrete value of `bedroom`, `bathroom`, and `den`.

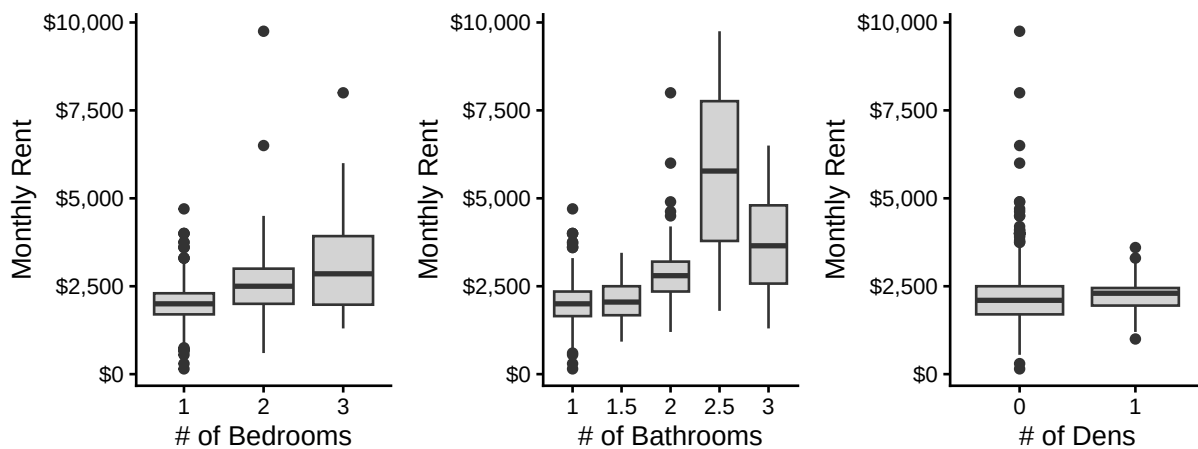
Frequencies of Room Counts by Room Type



TODO: comment on them (such as the unbalanced groups)

Below are boxplots of `price` split by `bedroom`, `bathroom`, and `den` counts in order to see how `price`'s distribution varies by each of these 3 apartment characteristics.

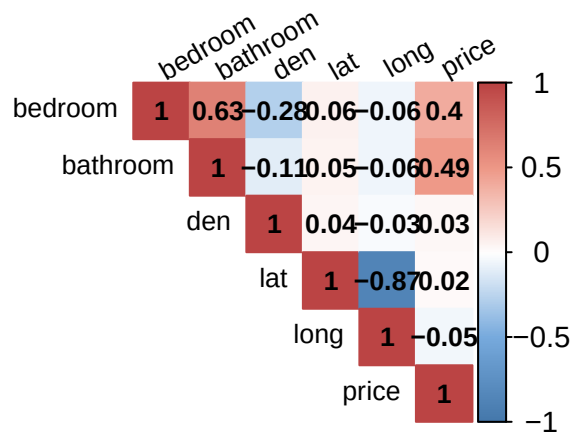
Distributions of Rent Prices by Room Counts



TODO: comment on them

TODO: number of observations of each 0/1 column

We're also curious about the correlations between the variables so we plot (the upper triangle of) a correlation matrix.



The variables with the strongest correlation with **price** are **bathroom** ($r = 0.47$) and **bedroom** ($r = 0.38$), though these are only moderately strong linear correlations. **den** is weakly negatively correlated with **price** at $r = -0.01$. Moreover, **bedroom** and **bathroom** are moderately correlated at $r = 0.50$, which may indicate that they're collinear, which may affect our modeling decisions later on.

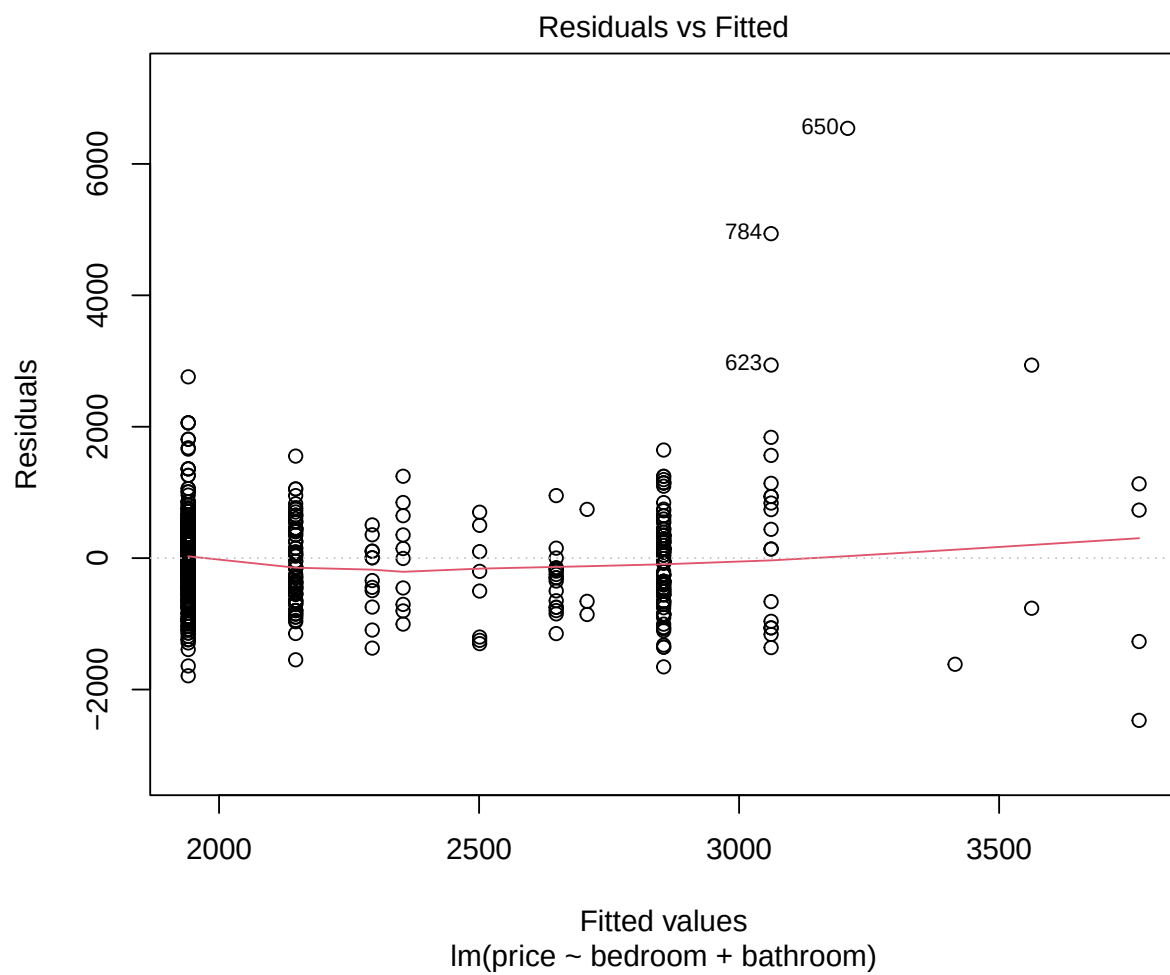
TODO: key take-aways of EDA

Models

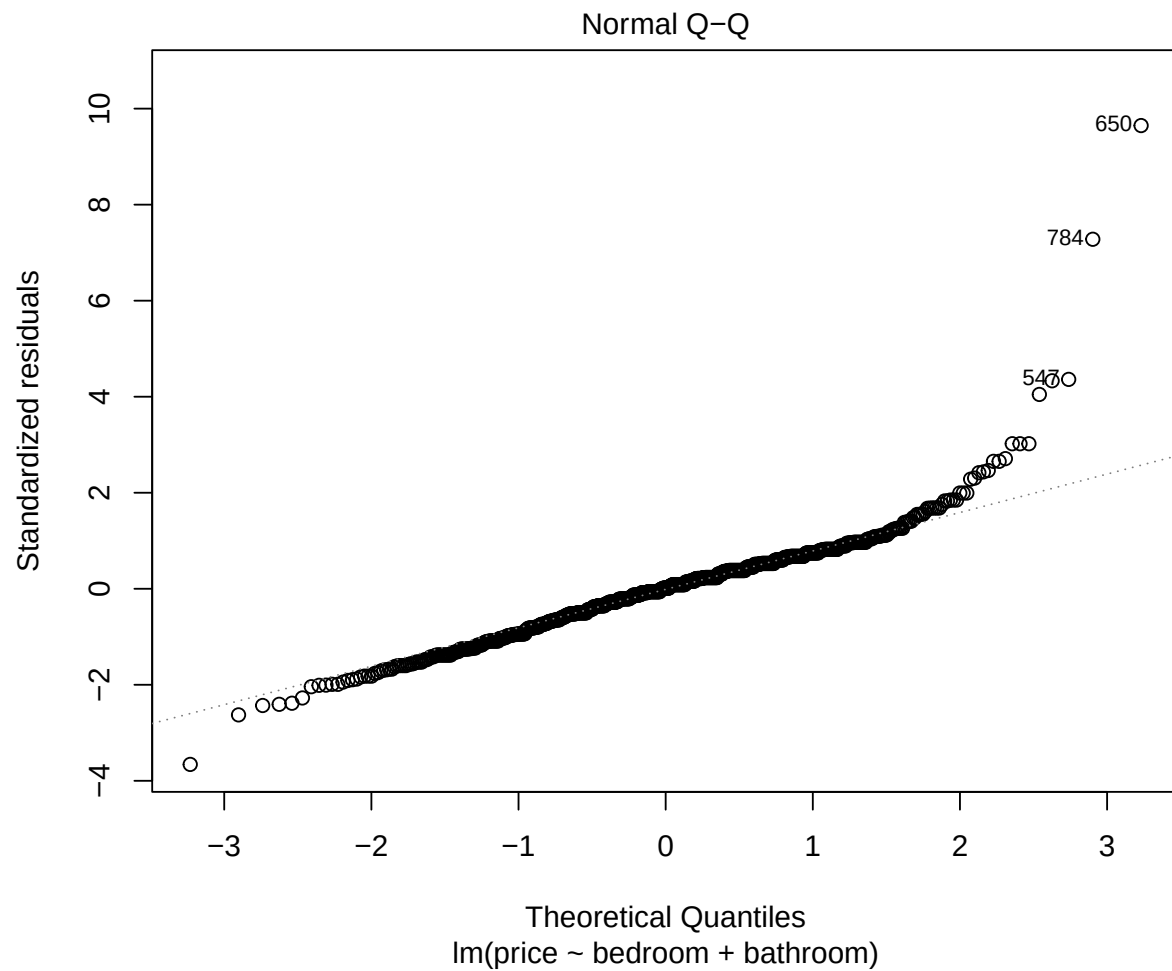
Additive Linear Regression

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1026.44	74.50	13.78	0	880.20	1172.68
bedroom	206.60	54.54	3.79	0	99.53	313.66
bathroom	707.66	70.73	10.01	0	568.83	846.49

TODO: commentary



TODO: commentary



Additive Logistic Regression

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-0.04	0.05	-0.85	0.40	-0.14	0.06
bedroom	0.05	0.04	1.30	0.19	-0.02	0.12
bathroom	0.36	0.05	7.44	0.00	0.27	0.46

TODO: commentary

TODO: maybe precision, accuracy, etc. as diagnostics

Conclusion

This section should provide overall conclusions, any limitations in the datasets and analyses, and validity of the conclusions.

Appendix

```
# Loading packages  
library(tidyverse)  
library(knitr)  
library(scales)  
library(patchwork)  
library(corrplot)  
library(broom)
```